Lecture #19 | Using API to retrieve data

SE377 Introduction to Big Data Analysis and Visualization (2017)

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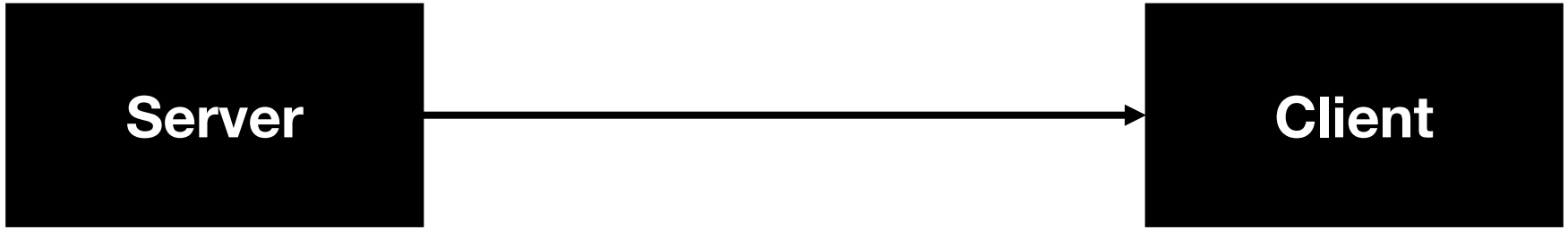
Today's topic

- Data representation on the web
  - JSON
  - XML
- More on HTTP
  - URL encoding for parameters
  - Session
- REST API
- Examples

Data representation on the web

- To exchange data between a server and a client, we need to have a common data representation that can be understood by both entities
  - Usually, the data exchanged has a form of hierarchical sequences or key-value pairs (e.g., `list` or `dict` in python)



- Two commonly used formats: XML and JSON

JavaScript Object Notation (JSON)

- Although JavaScript is in its name, JSON is a widely used format regardless of the implementation languages
- Example

```
{
    "note": {
        "to": "Tove",
        "from": "Jani",
        "heading": "Reminder",
        "body": "Don't forget me this weekend!”
    }
}
```

eXtensible Markup Language (XML)

- XML looks similar to html; it is composed of hierarchical tags, but with arbitrary tag names
- XML is not only used on the web, but also in the database domain
- Example

```
<?xml version="1.0" encoding="UTF-8"?>
<note>
  <to>Tove</to>
  <from>Jani</from>
  <heading>Reminder</heading>
  <body>Don't forget me this weekend!</body>
</note>
```

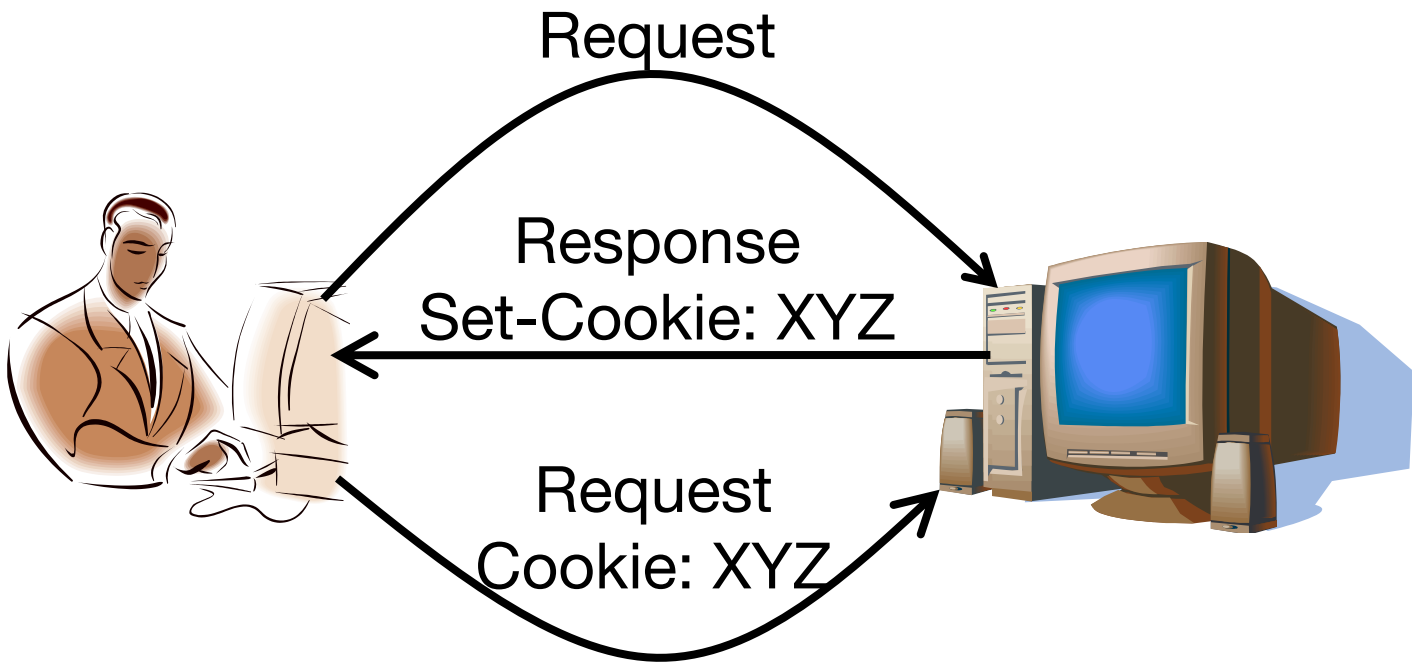
HTTP parameter encoding

- When we make a request with HTTP, we may specify options, similar to
  - Arguments of a function
  - Command line arguments of a process
- Arguments are passed in the headers or in URL
- To pass parameters encoded in the URL
  - Put '?' after the ordinary URL to indicate the start of parameters
  - Provide key-value pairs separated by '='
  - Put '&' between parameters
- Example

  http://prefix.domain:port/path/filename?key1=value1&key2=value2

Recap: providing states (or session)

- Client-side state maintenance
  - Client stores small(?) state on behalf of server
  - Client sends state in future requests to the server
- Can provide authentication



Use requests to provide sessions or parameters

```
url = 'https://www.mangoplate.com/top_lists/1268_daegu'
url_rest =
'https://stage.mangoplate.com/api/v2/web/top_lists/1268_daegu/restaurants
.js'

session = requests.session()
res1 = session.get(url)
params = {
    'language': 'kor',
    'start_index': '10',
    'request_count': '10'
}
res2 = session.get(url_rest, params=params)
res2_dict = json.loads(res2.text)
res2_dict
```

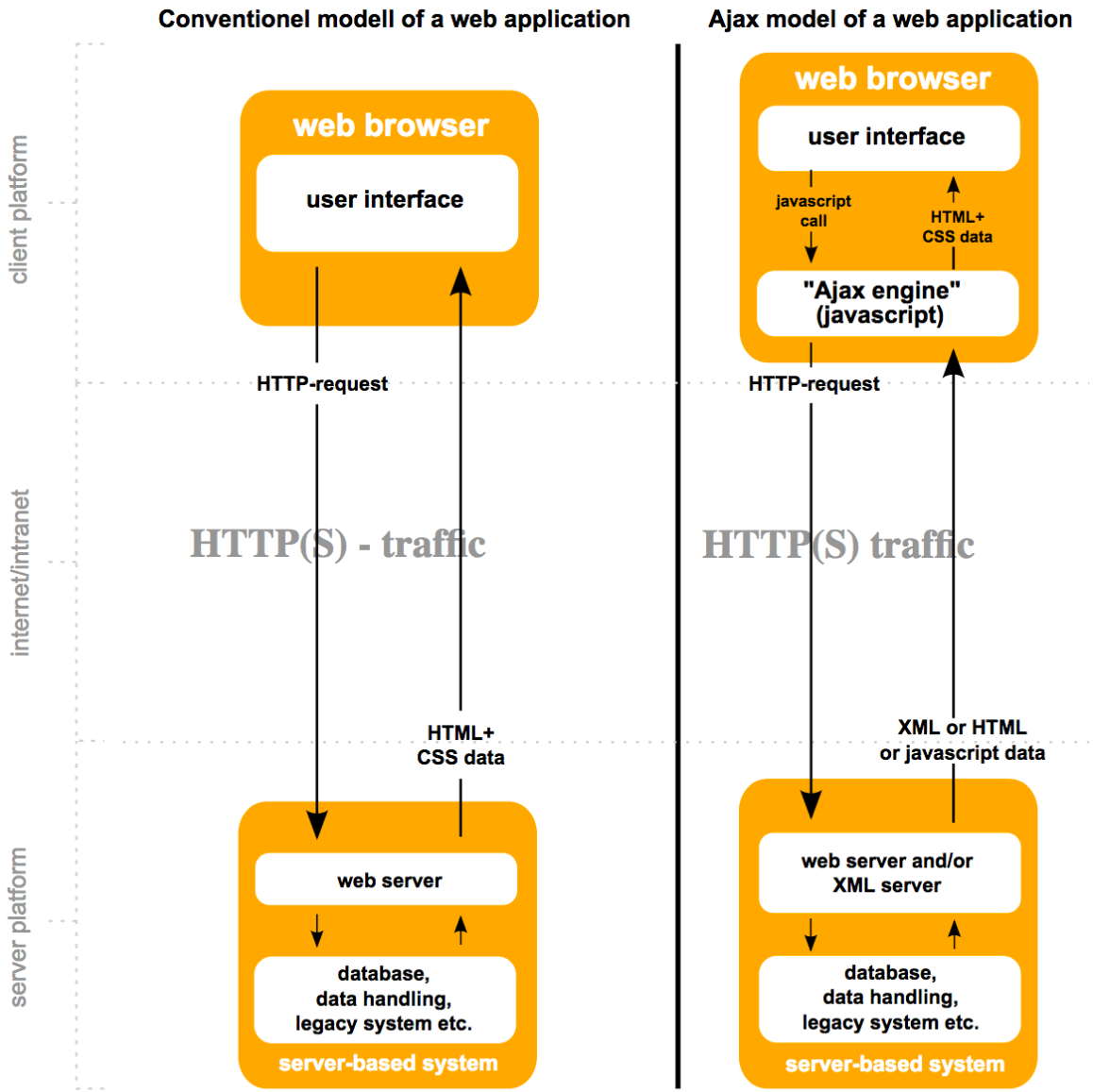
Asynchronous JavaScript and XML (AJAX)

- Asynchronous JavaScript and XML (AJAX) enables an asynchronous web application using
  - HTML (or XHTML) and CSS for presentation
  - The Document Object Model (DOM) for dynamic display of and interaction with data
  - JSON or XML for the interchange of data, and XSLT for its manipulation
  - The XMLHttpRequest object for asynchronous communication
  - JavaScript to bring these technologies together

Source: https://en.wikipedia.org/wiki/Ajax_(programming)

AJAX (cont.)

- Please stop using alien language!!!
- In a plain English (I am trying…)
- To show a new information (from the server),
  - Without AJAX, a new page should be loaded
  - With AJAX, new information can be presented in the current web page
    → much better responsiveness



CRUD: basic operations of data manipulation

- CRUD: create, read, update and delete
  - Basic operations of data manipulation, esp. for persistent memory (e.g., disk)

Operation	SQL	HTTP	REST
Create	INSERT	PUT/POST	POST
Read (Retrieve)	SELECT	GET	GET
Update (Modify)	UPDATE	PUT/POST/PATCH	PUT
Delete (Destroy)	DELETE	DELETE	DELETE

Source: https://en.wikipedia.org/wiki/Create,_read,_update_and_delete

Representational State Transfer (REST)

- REST or RESTful services provide interoperability over the Internet
- REST is NOT
  - Framework
  - Technology
  - Protocol
  - Standard specification
- REST is an architectural style
  - Guideline to design web service to provide performance, simplicity, scalability,mod

Additional materials

- JSON:
  - Official site: https://www.json.org/
  - JSON introduction: https://www.w3schools.com/js/js_json_intro.asp
- XML
  - XML core working group: https://www.w3.org/XML/
  - XML introduction: https://www.w3schools.com/xml/
  - Elements and Element Trees: http://effbot.org/zone/element.htm
- Articles
  - 인공지능 시대, 데이터 지능을 갖춘 전문가가 생존한다

ANY QUESTIONS?